

UNIT NO: 1 Core Components of PC/Laptop

❖ Introduction of Computer Systems and PC Development

Computer System: It is a complete working hardware and software setup that process data to perform specific task.

Computer: A **computer** is an electronic device that accepts data as input, processes it according to a set of instructions, stores it, and produces meaningful output.

Basic Functions of a Computer

1. Input
2. Processing
3. Storage
4. Output
5. Control

❖ Evolution of Personal Computers (PC)

Generation	Technology	Main Feature
First	Vacuum Tubes	Large size, high power consumption
Second	Transistors	Smaller and faster
Third	Integrated Circuits (IC)	More reliable
Fourth	Microprocessors	Birth of Personal Computers
Fifth	Artificial Intelligence	Smart and intelligent systems

❖ Types of Computers

A) Microcomputer or Personal Computer

- It is a small computer designed for use by a single user at a time.
- It uses a single chip (microprocessor) for its central processing unit (CPU).
- It can refer to any kind of small computer, such as a desktop computer, laptop computer, tablet, smart phone, or wearable.
- PC has low cost and smaller size.
- It is useful for individuals without intermediate computer operator
- The most common operating system such as Microsoft Window, Mac OS , UNIX, LINUX etc are used in PC.
- It can be capable to connect with other PC on LAN.
- Examples: Desktop PC, Laptop, Tablet.

Features:

- Low cost
- Small-sized computer.
- Designed for single-user operation.
- Easy to use

Types of personal computers

- ✓ **Desktop computer** – a personal computer that is designed to stay at one location and fits on or under a desk. It has a monitor, keyboard, mouse, and a tower (system unit).

- ✓ **Laptop computer** (or notebook) – A portable personal computer that is small enough to rest on the user's lap and can be powered by a battery.
- ✓ **Tablet** – A wireless touch screen PC that is slightly smaller and less weights than the average laptop.
- ✓ **Smart phones:** A mobile phone that performs many of the functions of a personal computer.

B) Mainframe Computer

- It is **Very large and powerful computer**.
- A **high-performance computer** used for **large information processing jobs**.
- The Processing power of the mainframe is time- shared between all of the users.
- It can **handle thousands of users**.
- Sometimes called "**big iron**".
- It requires greater **availability and security** than personal computer.
- In past, it is centralized instead of distributed computing.
- Users connect to the mainframe using terminals and submit their tasks for processing by the mainframe.(terminal has input and output but no processing power)
- It is used in banks, airlines, Research, Health care, large business and government organizations.



Features:

- High processing power
- Large storage capacity
- High reliability

C) Minicomputer

- It is Medium-sized computer.
- It is a mid-range computer that is intermediate in size, power, speed, storage capacity, etc., between a mainframe and a microcomputer.
- It supports multiple users simultaneously.
- It is used in small organizations.
- In recent years, minicomputers are often referred to as small or midsize servers

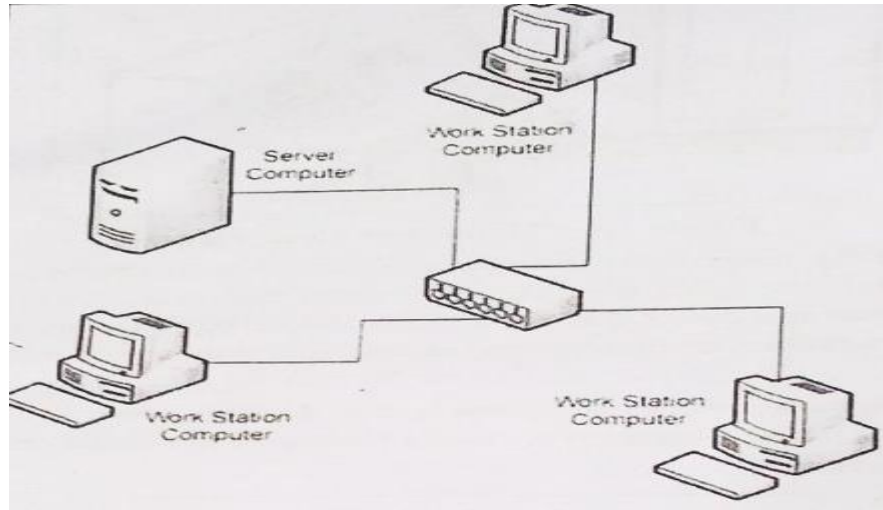


Examples:

- IBM AS/400

D) Workstation

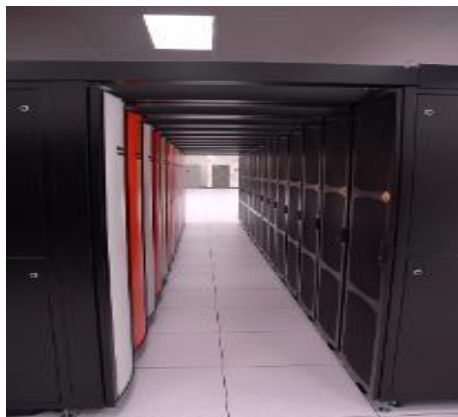
- It is high-performance computer for **technical work** like **3D rendering, CAD, video editing** etc.
- They contain **one or more microprocessor CPUs**.
- They may be used by a **single-user for applications** requiring **more power than** a typical PC.
- It **used by engineers, architects, and graphic designers**.
- In networking, any client computer connected to the network that access server resources may be called workstation.

**Features:**

- Powerful CPU
- Large RAM
- High-end graphics

E) Super computer

- Fastest and most powerful computer.
- It is a mainframe computer that has been optimized for **speed and processing power**.
- **Used for scientific research and weather forecasting.**
- It is useful for applications which involves **very large databases or require a great amount of computation**.
- **Most supercomputer is multiple computers that work parallel.**
- It is a powerful computer that can process large amounts of data and do a great amount of computation very quickly.
- IBM's supercomputer is 15 times faster than average PC and consists of 5800 processors.



Applications:

- Space research
- Climate prediction
- Nuclear simulations
- Weather forecasting

F) Cloud Computing System

- Computing resources like networking, server, storage, software etc. are provided over the Internet.
- Data and applications are stored on remote servers.

Advantages:

- Access from anywhere
- Cost-effective
- Easy backup

G) Embedded Systems

- Special-purpose computers built into devices.
- It has low power consumption.
- It is a single system within a larger device that is designed to perform one specific task.
- They are typically self-sufficient and require no user interface or human control to operate.

Examples:

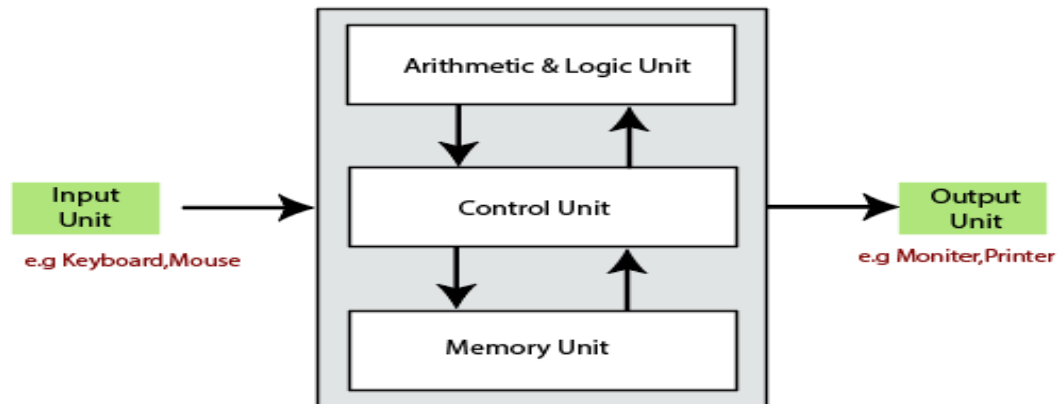
- Washing machine
- Smart TV
- ATM machine
- Car control system

❖ Differentiate Mainframe VS Minicomputer computer

Mainframe computer	Minicomputer computer
large size of disk is used.	Small size of disk is used.
It has high data handling capacity.	It has low data handling capacity.
Mainframe computers have large memory storage.	minicomputers have small or less memory storage than mainframe computer.
The processing speed of mainframe computer is faster than minicomputer.	While the processing speed of minicomputer is slower than mainframe computer.
They are more expensive.	It is not as expensive as mainframes.
It is more complex	It is less complex
They cannot be used as personal computers.	They can be used as personal computers.
It is More reliable	It is Less reliable
Mainframe computers support thousands or millions of users simultaneously.	Whereas minicomputers support hundreds of users at a time.
The first successful mainframe computer was invented by IBM.	The first microcomputer was invented by Bill Pentz.

❖ PC Block Diagram and System Components

Central Processing Unit (CPU)



- **CPU (Central Processing Unit)**
The CPU is called the **brain of the computer**.

Main Parts of CPU

A) ALU (Arithmetic Logic Unit)

Performs:

- Addition
- Subtraction
- Multiplication
- Division
- Logical operations (AND, OR, NOT)

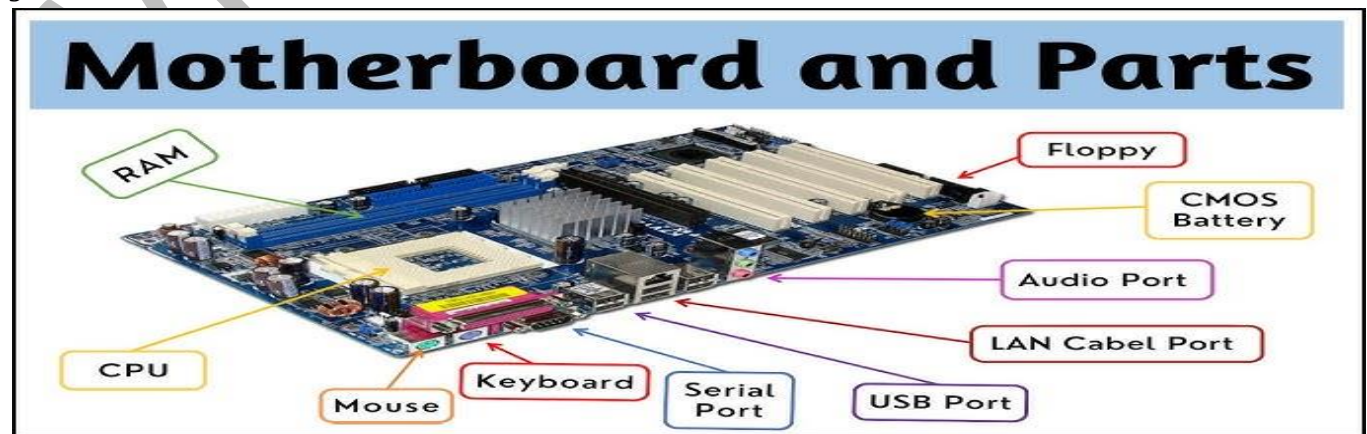
B) CU (Control Unit)

- Controls all computer operations.
- Coordinates input, processing, and output.
- Manages communication between hardware components.

- **Motherboard**

The motherboard is the **main circuit board** of a computer.

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Functions

- Connects all hardware components.
- Provides communication between CPU, RAM, storage, and peripherals.
- Supplies power to connected devices.

Components on Motherboard

- CPU Socket
- RAM Slots
- BIOS/UEFI Chip
- Expansion Slots
- SATA Connectors
- USB Ports
- Power Connectors

- **RAM (Random Access Memory)**

RAM is a **temporary memory** used while the computer is running.

Features

- Volatile memory
- Fast access speed
- Stores currently running programs

Types

- SRAM
- DRAM

Example: 8GB DDR4 RAM

- **ROM (Read Only Memory)**

ROM stores permanent instructions required to start the computer.

Features

- Non-volatile memory
- Data is not lost when power is off

Types

- PROM
- EPROM
- EEPROM

- **Storage Devices**

A) HDD (Hard Disk Drive)

- Uses magnetic disks.
- Large storage capacity.
- Lower cost.

Advantages

- Cheap
- High storage

Disadvantages

- Slower
- Mechanical parts can fail

B) SSD (Solid State Drive)

- Uses flash memory.
- No moving parts.

Advantages

- Faster
- More durable
- Lower power consumption

Disadvantages

- More expensive

Feature	HDD	SSD
Speed	Slow	Fast
Cost	Low	High
Durability	Less	More
Noise	Yes	No

- **Keyboard:** It is input device. An input device used to enter text and commands.

Types of Keys

- Alphabet Keys
- Numeric Keys
- Function Keys
- Control Keys
- Special Keys

- **Mouse**

A pointing input device.

Types

- Mechanical Mouse
- Optical Mouse
- Wireless Mouse

Main Buttons

- Left Button
- Right Button
- Scroll Wheel

- **Monitor**

An output device that displays information.

LCD (Liquid Crystal Display)

- Thin and lightweight
- Low power consumption

LED (Light Emitting Diode)

- Better brightness
- Better contrast
- Energy efficient

LCD	LED
Uses fluorescent backlight	Uses LED backlight
Lower brightness	Higher brightness
Less energy efficient	More energy efficient

- **SMPS (Switched Mode Power Supply)**

SMPS converts AC power into regulated DC power required by computer components.



Functions

- Converts 230V AC to DC voltages.
- Protects components from power fluctuations.
- Supplies power to motherboard, CPU, HDD, SSD, etc.

❖ Hardware, Software and Firmware

Hardware:

- Physical components of a computer.
- Parts of the computer “we can see and touch”.
- Computer Hardware is the physical part of the computer system, the machinery and equipment.
- Types of computer Hardware:
 - ✓ Input devices : keyboard, mouse
 - ✓ Output device: monitor, printer
 - ✓ Storage devices: CD, DVD, HDD
 - ✓ Internal components: Hardware device present in the system like CPU, mother board etc.

Software:

- A set of programs that instructs hardware.
- It is a collection of instructions, procedures and documentation that performs different tasks on a computer system.
- They are programs that tell the computer what to do.

Types

1. System Software : design to control the computer’s internal functionality
 - Operating System
 - Device Drivers
2. Application Software: performs specific tasks for users.

- MS Word
- Web Browser

Firmware:

- Firmware is a special type of software permanently stored in ROM or Flash Memory.
- It is a form of microcode or program embedded into hardware devices to help them operate effectively.
- Also referred as software for hardware.
- Firmware is low-level software often stored on electrically programmable memory devices.
- Firmware is software installed at the time of manufacturing any hardware, including hardware such as keyboards, hard drives, BIOS, graphic cards, and printers
- The firmware contains instruction programs to perform the basic functions of any hardware.
- **Examples:**
 - BIOS
 - Router Firmware
 - Printer Firmware

Difference Between Hardware, Software and Firmware

Hardware	Software	Firmware
Physical device	Program (Set of instructions)	Permanent software
Can be see and touched	Cannot be touched	Stored in ROM/Flash
Different types of hardware	Two types: System software and application software	No such categories of firmware
All physical components of computer is hardware	Software cannot be a firmware.	Firmware can be a software
Example: CPU	Example: Windows	Example: BIOS
Updating is more difficult than software and firmware.	Updating of software is usually easier.	Updating of firmware is usually difficult.

❖ BIOS, UEFI and CMOS**BIOS (Basic Input Output System):**

- BIOS is firmware that starts the computer and checks hardware before loading the operating system.
- It is a tiny piece of code that is stored on a special chip called EPROM (Erasable Programmable Read-Only Memory).
- It contains the necessary instructions for the computer to start and load the operating system.

Functions

- Power-On Self Test (POST)
- Detect hardware
- Load operating system
- Configure system settings

UEFI (Unified Extensible Firmware Interface):

- UEFI is the modern replacement for BIOS.
- Its main job is to keep important information about the device initialization which is stored on a special part of the disk.
- UEFI tackle the limitations of BIOS and shorten the system boot time.
- It makes the computer start faster.
- It bridges the gap between the computer's hardware and operating system, allowing for better compatibility, security, and functionality.

Advantages

- Faster boot time
- Supports larger hard drives
- Better security
- Graphical interface

CMOS (Complementary Metal Oxide Semiconductor):

- CMOS is a small memory chip that stores BIOS settings.
- The CMOS chip is a compact, low-power device that holds crucial computer system setup data, such as the current date and time, hard drive type, and memory installed.
- The CMOS chip is typically found near the BIOS chip, which is in charge of initializing and testing hardware components during startup
- The CMOS chip is frequently used to store the current date and time.
- The operating system and other software applications use this information to precisely timestamp files and events.

It Stores

- Date and time
- Boot sequence
- Hardware configuration

A CMOS battery maintains these settings when the computer is turned off.

❖ Components of a laptop:

A laptop is a portable computer that integrates all essential hardware components into a compact device. These components work together to process data, store information, display output, and communicate with external devices.

- **CPU (Central Processing Unit):** The CPU is known as the **brain of the computer**. It performs calculations, executes instructions, and controls all computer operations.

Main Functions

- Executes programs and instructions.

- Performs arithmetic and logical operations.
- Controls the activities of all hardware components.

Components of CPU**a) Arithmetic Logic Unit (ALU)**

- Performs arithmetic operations (+, −, ×, ÷).
- Performs logical operations (AND, OR, NOT, comparison).

b) Control Unit (CU)

- Controls the execution of instructions.
- Coordinates communication between CPU, memory, and peripherals.

- **Memory (RAM):** RAM (Random Access Memory) is a temporary storage area used by the CPU while programs are running.

Features

- Volatile memory (data is lost when power is OFF).
- Provides fast access to data.
- Improves system performance.

Functions

- Stores currently running programs.
- Stores temporary data for processing.

Common Sizes

- 4 GB
- 8 GB
- 16 GB
- 32 GB

- **Expansion Slots:**

- An expansion slot is a port on a motherboard that accepts an expansion card
- Expansion slots allow additional hardware components to be connected to **the laptop**.

Purpose

- Increase functionality.
- Upgrade system capabilities.

Examples

- M.2 Slot for SSD
- Memory Slot (RAM Expansion)
- PCIe Expansion Interface

Benefits

- Storage upgrades
- RAM upgrades
- Wireless card upgrades

- **Power Supply (Battery):** The battery supplies electrical power to the laptop when it is not connected to AC power.

Types

- Lithium-Ion (Li-Ion)
- Lithium-Polymer (Li-Po)

Functions

- Provides portability.
- Supplies power during power failures.

Characteristics

- Rechargeable
- Lightweight
- Long battery life

- **Video Display Controller:** The Video Display Controller (Graphics Controller/GPU) generates images and videos for display on the screen.

Functions

- Processes graphics.
- Renders images and videos.
- Supports gaming and multimedia applications.

Examples

- Intel UHD Graphics
- NVIDIA GeForce
- AMD Radeon

- **Display:** The display is the screen that shows text, images, videos, and graphical output.

Types

- LCD (Liquid Crystal Display)
- LED (Light Emitting Diode)

Features

- Screen Size: 13", 14", 15.6", 17"
- Resolution: HD, Full HD, 2K, 4K
- Brightness and Color Quality

Function: Converts digital output into visual information.

- **Removable Media Drives :** These devices allow users to read and write data on removable storage media.

Examples

- USB Flash Drive
- Memory Card Reader
- External DVD Drive

Advantages

- Easy data transfer
- Backup storage
- Portable

- **Internal Storage :** Internal storage permanently stores the operating system, applications, and user data.

Types

Hard Disk Drive (HDD)

- Magnetic storage technology.
- Large capacity.
- Lower cost.

Solid State Drive (SSD)

- Flash memory technology.
- Faster than HDD.
- More reliable.

- **Input Devices:** Input devices are used to enter data and instructions into the laptop.

Common Input Devices :

Keyboard

- Used for typing.
- Contains alphanumeric and function keys.

Touchpad

- Used as a pointing device.
- Controls cursor movement.

Touchscreen (Optional)

- Allows direct interaction with display.

Microphone

- Inputs voice and audio.

- **Ports:** Ports provide physical connections between the laptop and external devices.

Common Laptop Ports

USB Port: Connects keyboard, mouse, pen drive, etc.

HDMI Port :Connects external monitors and projectors.

Audio Jack : Connects headphones and microphones.

Ethernet Port :Wired network connection.

Type-C Port : Data transfer, charging, and video output.

- **Cooling System (Fan / Heat Sink):** The cooling system removes excess heat generated by internal components.

Components

Fan :Circulates air inside the laptop.

Heat Sink :Absorbs and dissipates heat from CPU and GPU.

Functions

- Prevents overheating.
- Improves performance.
- Increases hardware lifespan.

Benefits

- Stable operation
- Better efficiency
- Reduced hardware damage

• **Speakers and Webcam**

Speakers provide audio output from the laptop.

Uses

- Music
- Videos
- Online meetings
- Gaming

A webcam is a built-in digital camera used for capturing images and video.

Uses

- Video conferencing
- Online classes
- Video recording

Examples

- Zoom Meetings
- Google Meet
- Microsoft Teams

• **Wireless Module (Wi-Fi & Bluetooth) :Wi-Fi enables wireless internet and network connectivity.**

Functions

- Internet access
- Wireless networking

Advantages

- Mobility
- No cables required

Bluetooth : Bluetooth is a short-range wireless communication technology.

Functions

- Connect wireless mouse
- Connect keyboard
- Connect headphones
- Transfer files

Advantages

- Low power consumption
- Easy device pairing

❖ Front & back panel overview –

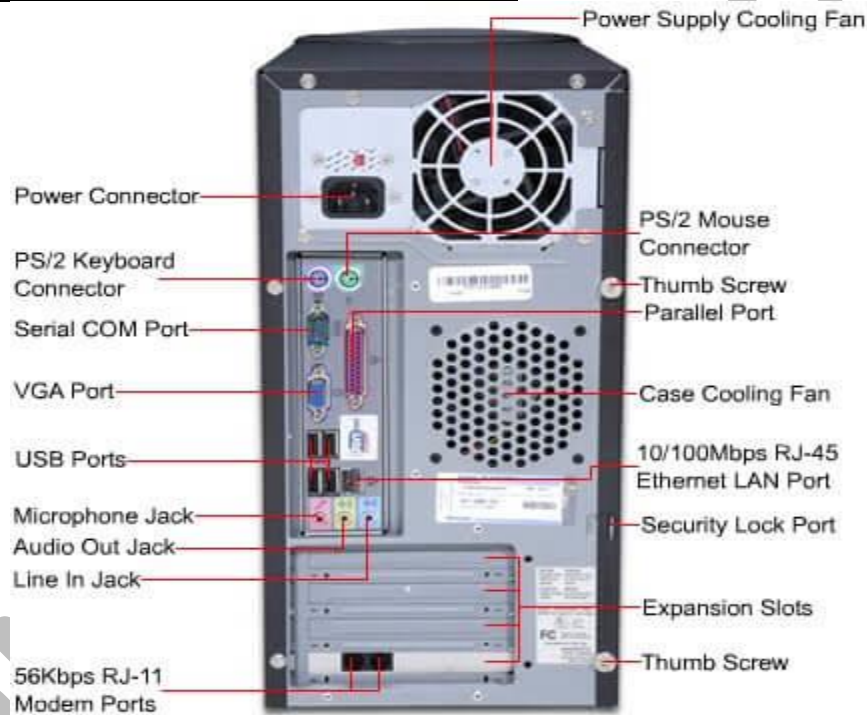
- A **computer port** is an interface used to connect external devices such as keyboards, mouse, monitors, printers, speakers, storage devices, and networks to a computer.
- The **front panel** provides ports that are frequently used by users, while the **back panel** contains permanent connections for major peripherals.
- The **front panel** of a desktop computer cabinet usually contains:



Component	Purpose
Power Button	Turns the computer ON or OFF
Reset Button	Restarts the computer without turning off the power supply
Power LED	Indicates whether the computer is ON
HDD LED	Blinks while the hard disk is reading or writing data
USB Ports	Connect USB devices
USB Type-C Port	Connects modern high-speed devices
Audio Jack	Connects headphones, speakers, and microphones
Memory Card Reader	Reads SD cards (optional)
Optical Drive	Reads CD/DVD (optional)

- **Back Panel Overview:** The back panel contains motherboard I/O ports.

Component	Function
Power Connector	Connects SMPS to AC supply
USB Ports	Connect keyboard, mouse, printer, etc.
HDMI Port	Connect monitor or TV
VGA Port	Connect older monitors
RJ45 Port	Connect LAN cable
Audio Ports	Connect speakers and microphones
PS/2 Port	Connect old keyboard or mouse
DisplayPort	Connect high-resolution monitors
Thunderbolt Port	High-speed data, video, and charging



CPU Back Panel

Ports: A port is a physical slot or interface on the computer's motherboard or expansion card.

- **USB(Universal Serial Bus) Port:** A USB (Universal Serial Bus) port is a common interface used to connect peripheral devices to a computer.



- Used for connecting keyboards, mice, printers, and storage devices

- Supports data transfer and power supply
- Plug-and-play, easy to use
- Available in different versions like USB 2.0, USB 3.0, and USB-C

Advantages

- High-speed communication
- Easy installation
- Supports many devices
- Portable
- Provides power and data together

- **HDMI Port : An HDMI (High-Definition Multimedia Interface)** port is used to transmit audio and video signals.



- Supports high-definition video and audio
- Used to connect monitors, TVs, and projectors
- Carries both audio and video through a single cable
- Common in modern multimedia devices

Applications

- TV
- Monitor
- Projector
- Gaming Console
- Laptop
- Desktop PC

Advantages

- Excellent picture quality
- Excellent sound quality
- Easy connection
- One cable for audio and video

- **Ethernet (RJ-45) Port:** An Ethernet port is used to connect a device to a wired network.



- Provides reliable and high-speed network connectivity
- Commonly used in LAN environments
- Uses Ethernet cables (Cat5, Cat6, etc.)
- More stable than wireless connections

Applications

- Office Network
- Internet Connection
- Switch
- Router
- Server

Advantages

- Reliable
- Faster than Wi-Fi
- Secure
- Low latency

- **Audio Ports :** Audio ports are interfaces used for sound input and output.



- Used to connect headphones, speakers, and microphones
- Usually color-coded for easy identification
- Support analog audio signals
- Found in computers, laptops, and multimedia devices

Applications

- Speakers
- Headphones
- Earphones
- Microphones

Advantages

- Easy to use
- Low cost
- Widely compatible

- **VGA / DVI / DisplayPort :** These ports are used for video output from a computer to a display device. The **VGA Port** is an analog display interface used to connect monitors and projectors.



- VGA is an analog video interface
- DVI supports digital video signals
- Display Port supports high-resolution and high-refresh-rate displays
- Commonly used to connect monitors
- 15-pin connector
- No audio support
- Mostly used in older computers

Applications

- CRT Monitor
- LCD Monitor
- Projector

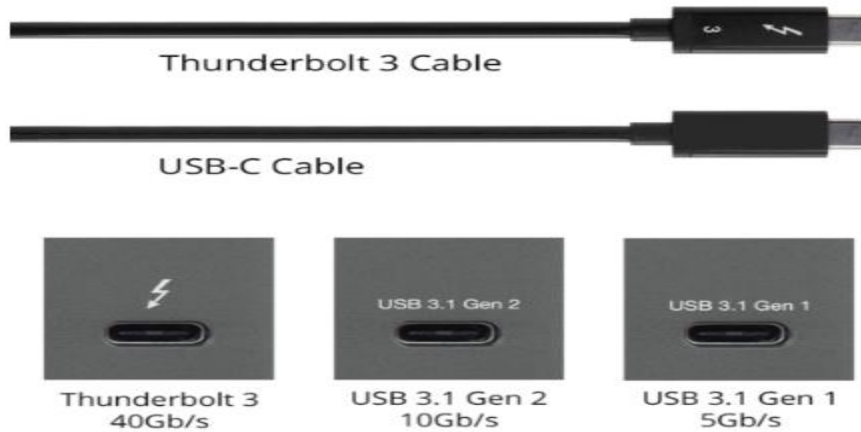
Advantages

- Simple
- Widely available in older systems

- **PS/2 Port:** A PS/2 port is an older interface used to connect basic input devices.



- Mainly used for keyboards and mice
 - Circular connector with pin configuration
 - Found in older computer systems
 - Provides direct communication with the device
- **Thunderbolt Port :** Thunderbolt is a **high-speed interface** developed by Intel that supports **data transfer, video output, audio, networking, and charging** through a single port.



- offers significantly faster data transfer speeds
- It offers the fastest data transfer speeds, supporting up to 40 gigabit per second (Gbps)
- It supports both data and display protocols
- allowing to connect a wide range of devices such as external hard drives, monitors, audio interfaces, and even docking stations

Applications

- External SSD
- Docking Station
- High-resolution Monitor
- External GPU
- Professional Video Editing

Advantages

- Very high speed
- Multi-purpose
- Daisy chain devices
- Fast charging

Connectors: A connector is the physical plug attached to a cable that is inserted into a port.

- **USB Connector**

- Connecting peripherals like mice, keyboards, and external drives.
- Transfers data and provides electrical power to devices.
- Ranges from older USB-A to the modern, reversible USB-C

Types

USB Type-A

- Rectangular
- Most common
- Desktop and Laptop

USB Type-B

- Square shape
- Printers
- Scanners

Mini USB

- Older cameras

Micro USB

- Android phones (older)

USB Type-C

- Reversible
- Fast charging
- Latest smartphones
- Laptops

• HDMI Connector

- Transmitting high-definition video and audio.
- Displays, TVs, gaming consoles, and projectors.
- Carries both uncompressed video and multi-channel audio signal

Features

- Carries digital audio
- Carries digital video
- High-definition support

Type	Usage
Type A	TV, Monitor, Laptop
Type C (Mini HDMI)	Camera
Type D (Micro HDMI)	Tablets, Mobile Devices

• RJ45 Connector

- The RJ45 connector looks like a large phone jack.
- It has 8 pins inside that correspond with the 8 wires in an Ethernet cable.
- RJ45 is the standard connector for Ethernet networks.
- Ethernet port or network connector.
- Connects computers to routers or switches for stable internet
- **Uses**
 - LAN Cable
 - Router
 - Switch
 - Computer

• Audio Jack Connector

- Connecting analog audio devices like headphones and microphones.
- Most frequently found in the standard 3.5mm size.
- Carries stereo sound and sometimes input for a mic
- **Types**
 - 3.5 mm Jack : Most common.
 - 6.35 mm Jack: Professional audio equipment.
 - TRS Connector : Stereo audio.
 - TRRS Connector: Audio + Microphone.
- **Applications**

- Headphones
- Speakers
- Microphones

• VGA Connector

- Legacy video connection for older monitors and projectors.
- It is a **15-pin D-Sub connector** used for analog video transmission.
- Transmits analog video signals only (no audio).
- Notable for its 15-pin layout and blue color.
- **Features**
 - Blue connector
 - Screw locking
 - Analog signal
- **Applications**
 - Monitor
 - Projector

• Thunderbolt Connector

- High-speed data, video output, and power delivery.
- Modern versions (Thunderbolt 3 and 4) use the USB-C shape.
- Offers massive bandwidth for external graphics cards and dual 4K monitors
- **Types**
 - Thunderbolt 1 & 2 : Mini DisplayPort connector.
 - Thunderbolt 3 & 4 :USB Type-C connector.
- **Features**
 - 40 Gbps speed
 - Video + Audio + Data
 - Charging support External GPU support
 -



❖ **Prepare the latest PC/laptop specifications**

Processor	11th, 12th or 13th Gen Intel Core i5, i7 or i9 Processor, Apple Silicon M1, M1 Pro, M1 Max, M1 Ultra or M2 Processor (CPU)
Operating system	• Windows 10 or Windows 11 Home/Pro (64-bit) • Ubuntu Linux 24.04 LTS (for developers) • macOS Sequoia (Apple laptops)
RAM	8-16 GB of RAM • Recommended: 32 GB for gaming, programming, video editing, AI applications
Storage	240 GB solid state drive, or larger.
Graphics Unit (GPU)	Integrated or discrete graphics processor capable of 1440 X 900 resolution, or better (1920 X 1080 or 1200 ideal).
Display	Size: 14", 15.6", or 16" • Resolution: Full HD (1920 × 1080) or 2.5K/3K • Anti-glare coating
Mouse	Built-in or external track pad, wireless and/or USB, 2-button, optical mouse
Sound	Built-in audio or soundcard, and speakers
Headphones	Headphones or ear buds, with built-in microphone
Webcam	Either external USB device or built-in
Ports	• USB Type-C (USB4/Thunderbolt 4 or 5) • USB Type-A 3.2 • HDMI 2.1 • Audio Combo Jack (3.5 mm) • SD/MicroSD Card Reader (optional) • Ethernet (RJ-45) on some models
Connectivity	Wi-Fi 6E or Wi-Fi 7 Bluetooth 5.3 or 5.4
Backup device/scheme	External USB or Thunderbolt drive with Time Machine (Mac) or Windows backup software and/or cloud backup service/subscription.

❖ **Compare between Port and Connectors.**

Port	Connectors
1) Sockets on devices where cables plugin	1) Plugs on the end of cables that connect to ports
2) Allow network connections	2) Physically join devices over networks
3) Fixed on devices	3) Can be removed and changed
4) Standard shapes and sizes	4) Vary based on port compatibility
5) Examples: Ethernet, USB, HDMI	5) Examples: RJ45, USB-A, HDMI